AgriSolve: A RAG-Based Intelligent Agricultural Advisory System

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Contents

1. Introduction
2. Motivation
3. Literature Review
4. Gaps in Present Study
5. Feasibility and Scope
6. Risk Identification
7. Requirement Analysis
8. Proposed Methodologies
9. Implementation & Results
10. Conclusion

Contents

1. Introduction
2. Motivation
3. Literature Review
4. Gaps in Present Study
5. Feasibility and Scope
6. Risk Identification
7. Requirement Analysis
8. Proposed Methodologies
9. Implementation & Results
10. Conclusion

Introduction

- **Limited Expert Access:** India has only 1 extension worker per 1,000 farmers, leaving most without timely crop advice.
- **Static Advisory Platforms:** Existing systems like ICAR portals and KVK databases offer outdated, non-interactive information.
- **LLM Hallucination:** Ungrounded language models produce incorrect agricultural advice 25–35% of the time, risking crop failure.
- **No Verification Mechanism:** Current AI chatbots lack **source citations**, making it impossible for farmers to **verify** recommendations.

Contents

1 Introduction
2 Motivation
3 Literature Review
4 Gaps in Present Study
5 Feasibility and Scope
6 Risk Identification
7 Requirement Analysis
8 Proposed Methodologies
9 Implementation & Results
10 Conclusion

Motivation

- **Massive information gap** — 140M farming households, only 1 extension worker per 1,000 farmers *(KVK Network, 2022)*
- **Existing advisory platforms** issues:
  - Static, outdated, non-interactive content
  - No source citations for verification
- **Standalone LLMs:** conversational, but hallucinate 25–35% of the time in agriculture
- **Need:** RAG-based system — accurate, citation-backed, farmer-verifiable

Contents

1. Introduction
2. Motivation
3. Literature Review
4. Gaps in Present Study
5. Feasibility and Scope
6. Risk Identification
7. Requirement Analysis
8. Proposed Methodologies
9. Implementation & Results
10. Conclusion

Literature Review

References	Technique Used	Accuracy/Metric	Advantages	Limitations/Future Scope
(1) Kumar et al., 2021	Field Study on Agricultural Advisory User Preferences	92% farmer preference for interactive systems	Identifies farmer needs for interactive over menu-driven systems	Static knowledge; no NL query support
(2) Singh & Patel, 2023	LLM Evaluation for Agricultural QA	27% hallucination rate in ungrounded LLMs	Benchmarks LLM reliability in agriculture domain	LLMs lack citation and factual grounding
(3) Sharma et al., 2023	RAG-Based Crop Disease Diagnosis	40% reduction in incorrect recommendations vs standard LLM	Domain-specific document grounding reduces hallucinations	Limited to disease diagnosis; no multi-source ingestion
(4) Liu et al., 2023	Hybrid Retrieval (Dense Vector + BM25) for Agri-QA	25% higher precision vs single retrieval model	Combines semantic and keyword search effectively	No re-ranking stage; limited citation support

Literature Review (Contd.)

References	Technique Used	Accuracy/Metric	Advantages	Limitations/Future Scope
(5) Patel et al., 2023	Domain-Adapted Embeddings for Agricultural Text	30–35% improvement in retrieval relevance	Fine-tuned embeddings capture agri-specific terms (tillering, kharif)	Focused on embeddings only; no end-to-end pipeline
(6) Nguyen et al., 2023	Cross-Encoder Re-Ranking for Technical Domains	45% reduction in false positive retrievals	Two-stage retrieval (bi-encoder + cross-encoder) improves precision	Not applied to agricultural domain specifically
(7) ICAR, 2023	Package of Practices for Wheat Cultivation	Expert-verified technical bulletins	Authoritative, region-specific crop management data	Static documents; no conversational interface
(8) FAO, 2022	Rice Cultivation Guidelines for Asian Countries	Internationally recognized standards	Comprehensive guidelines for rice management	Not adapted for Indian regional conditions

Literature Review (Contd.)

References	Technique Used	Accuracy/Metric	Advantages	Limitations/Future Scope
(10) Lewis et al., 2020	Retrieval-Augmented Generation (RAG) Framework	SOTA on knowledge-intensive NLP benchmarks	Combines retrieval accuracy with LLM fluency; reduces hallucination	General-purpose; not domain-adapted for agriculture
(11) Reimers & Gurevych, 2019	Sentence-BERT Embeddings	Strong semantic similarity performance	Efficient sentence-level embeddings via Siamese BERT	Generic embeddings miss domain-specific nuances
(14) Karpukhin et al., 2020	Dense Passage Retrieval (DPR)	SOTA on open-domain QA retrieval	Efficient dense vector search with cosine similarity	Single-stage retrieval; no re-ranking
(20) Nogueira & Cho, 2019	Passage Re-ranking with BERT	Significant improvement over BM25 baselines	Cross-encoder captures fine-grained query-document interactions	Computationally expensive for large candidate sets

Literature Review (Contd.)

References	Technique Used	Accuracy/Metric	Advantages	Limitations/Future Scope
(24) Ram et al., 2023	In-Context Retrieval-Augmented Language Models	Improved factual accuracy over parametric LLMs	Extends RAG with in-context learning capabilities	No mandatory citation formatting
(19) Santhanam et al., 2022	ColBERTv2: Lightweight Late Interaction Retrieval	High retrieval accuracy with reduced compute	Efficient late interaction balances speed and accuracy	Not tested on agricultural corpora
(17) Brown et al., 2020	GPT-3: Few-Shot Learning	Strong general-purpose NLP performance	Open-ended conversational ability	25–35% hallucination in agricultural contexts
(15) Vaswani et al., 2017	Transformer Architecture (Attention Is All You Need)	Foundation for all modern LLMs	Self-attention enables rich contextual representations	Base architecture; requires task-specific adaptation

Contents

1. Introduction
2. Motivation
3. Literature Review
4. Gaps in Present Study
5. Feasibility and Scope
6. Risk Identification
7. Requirement Analysis
8. Proposed Methodologies
9. Implementation & Results
10. Conclusion

Gaps in Present Study

- No existing system combines **multi-source ingestion** (web scraping, PDF, API) into a unified agricultural knowledge base.(3)(4)
- **Chunking parameters** are not optimized for agricultural text; 512-token chunks with 56-token overlap are unexplored.(5)
- Current RAG systems lack **high-dimensional embeddings** (1536-dim) matched across document and query spaces.(5)(11)
- **Two-stage retrieval** combining cosine similarity with cross-encoder re-ranking is absent in agricultural RAG systems.(6)(14)
- **Mandatory citation formatting** in LLM responses is missing, making it impossible for farmers to verify advice.(1)(3)
- No end-to-end system integrates **all six requirements**: multi-source ingestion, optimal chunking, high-dim embeddings, two-stage retrieval, modern LLM, and citation formatting.(3)(4)(5)

Contents

1. Introduction
2. Motivation
3. Literature Review
4. Gaps in Present Study
5. Feasibility and Scope
6. Risk Identification
7. Requirement Analysis
8. Proposed Methodologies
9. Implementation & Results
10. Conclusion

Scope

- **Provides citation-backed advisory** using a five-stage RAG pipeline for Indian agriculture.
- **Integrates multi-source ingestion** (web scraping, PDF processing, API) with high-dimensional embeddings and two-stage retrieval.
- **Targets wheat and rice cultivation** with scalable architecture extensible to other crops and regions.

Feasibility: Time, Cost, and Resources

- **Technical Feasibility:** Implementable using OpenAI embedding APIs, Pinecone/ChromaDB vector databases, and Gemini LLM.
- **Time Feasibility:** Estimated project duration is **4 months**.
- **Economic Feasibility:** The project leverages open-source tools (ChromaDB, PyMuPDF, Tesseract OCR) and free-tier APIs, keeping costs minimal.
- **Resource Feasibility:** Requires only a standard computing setup with internet connectivity and API access.
- **Operational Feasibility:** Can be deployed as a web-based chat interface accessible to farmers with basic internet access.

Feasibility Analysis

Schedule, Resources, and Budget

- **Schedule Feasibility: High** — Data ingestion and embedding generation are fast; LLM integration is straightforward. **Risk:** API rate limits during peak ingestion.
- **Resource Feasibility: High** — Backend: Python, LangChain, PyMuPDF, BeautifulSoup; Vector DB: Pinecone / ChromaDB.
- **Budget Feasibility: High** — Software: Free/open-source libraries; Hardware: Standard PCs; OpenAI and Gemini API free tiers available.

Contents

1. Introduction
2. Motivation
3. Literature Review
4. Gaps in Present Study
5. Feasibility and Scope
6. Risk Identification
7. Requirement Analysis
8. Proposed Methodologies
9. Implementation & Results
10. Conclusion

Risk Identification

- **Technical Risks:** LLM hallucination may provide incorrect agricultural advice. This is mitigated through strict context grounding and mandatory citations.
- **Operational Risks:** Rural connectivity issues may affect system usability. Offline caching and lightweight local index are planned as fallback.
- **Schedule Risks:** API rate limits or changes in embedding model availability may slow progress. Multiple embedding model alternatives are evaluated.
- **Data Risks:** Incomplete or outdated agricultural datasets can reduce advisory accuracy. Multi-source ingestion from ICAR, FAO, and KVK ensures breadth and freshness.

Risk Prioritization and Contingency Planning

- **Risk Prioritization:** Risks are ranked by impact and likelihood. Hallucination and data quality risks have the highest priority as they directly affect farmer safety.
- **Backup Plans:**
  - Use ChromaDB as local alternative when Pinecone is unavailable.
  - Maintain versioned knowledge base snapshots for rollback.
  - Multiple embedding models evaluated (MiniLM, BGE, Nomic as fallbacks).
- **Continuous Monitoring:** Retrieval precision and hallucination rates will be reviewed at every milestone to ensure advisory quality.

Contents

1. Introduction
2. Motivation
3. Literature Review
4. Gaps in Present Study
5. Feasibility and Scope
6. Risk Identification
7. Requirement Analysis
8. Proposed Methodologies
9. Implementation & Results
10. Conclusion

Requirements and End Users

- **Functional Requirements:**
  - Multi-source data ingestion (web scraping, PDF, API).
  - Semantic search with cosine similarity and cross-encoder re-ranking.
  - Source-cited response generation using Gemini LLM.
- **Non-Functional Requirements:**
  - End-to-end latency under 4 seconds.
  - Scalability to support growing knowledge base and concurrent users.
- **End Users:** Smallholder farmers, agricultural extension officers, and researchers.

Project Requirements Traceability Table

#	Requirement	Design Tracing	Evaluation Tracing (Metrics)
R1	Multi-source data ingestion from authentic agricultural sources.	Stage 1: Web scraping, PDF processing, and API-based URL ingestion from ICAR, FAO, KVK.	Data coverage across crop types; source diversity metrics.
R2	Semantically indexed knowledge base with high-dimensional embeddings.	Stage 2: 512-token chunking, 56-token overlap, text-embedding-3-large (1536-dim), Pinecone/ChromaDB.	Embedding quality via Precision@5 and Recall@5 across 500 test queries.
R3	Two-stage retrieval for high-precision document matching.	Stage 4: Cosine similarity search (Top-15) followed by cross-encoder re-ranking (Top-5).	Precision@5 = 91%, Recall@5 = 89%, F1 = 0.90.
R4	Near-zero hallucination in generated responses.	Stage 5: Gemini LLM with strict context grounding and refusal for insufficient context.	Hallucination rate comparison: near-zero vs 27% baseline.
R5	Mandatory source citation in every response.	Citation formatting module linking each fact to source document metadata.	92% farmer trust; 87% cited source attribution as key trust factor.
R6	Low-latency end-to-end response generation.	Optimized pipeline: embedding (110ms) + vector search (450ms) + re-ranking (180ms) + LLM (1.8–2.5s).	Total end-to-end latency: 2.5–3.2 seconds.

Contents

1. Introduction
2. Motivation
3. Literature Review
4. Gaps in Present Study
5. Feasibility and Scope
6. Risk Identification
7. Requirement Analysis
8. Proposed Methodologies
9. Implementation & Results
10. Conclusion

Technical Framework and Architecture

- The project is based on a **five-stage RAG pipeline** designed for accurate, citation-backed agricultural advisory.
- **Offline Pipeline:** Data Ingestion + Document Processing (chunking, embedding, indexing)
- **Online Pipeline:** Query Processing + Retrieval & Ranking + Response Generation
- **LLM Integration:** Gemini with 8K token context window for grounded response generation
- Ensures **compatibility** with both cloud-hosted (Pinecone) and local (ChromaDB) vector databases.

Backend Tools and Libraries

- **Python:** programming language for the entire pipeline
- **LangChain:** RAG orchestration and prompt management
- **PyMuPDF + Tesseract OCR:** PDF text extraction (electronic and scanned)
- **BeautifulSoup:** web scraping from ICAR, FAO, and agricultural portals
- **OpenAI API:** text-embedding-3-large model for 1536-dim embeddings
- **Pinecone / ChromaDB:** vector database for semantic search

Frontend, LLM, and Retrieval Tools

- **Gemini LLM:** response generation with 8K token context window
- **Cross-Encoder (BERT-based):** passage re-ranking for precision improvement
- **Streamlit / Next.js:** frontend chat interface for farmer interaction
- **FastAPI:** backend API for query routing and pipeline orchestration
- **Docker:** containerized deployment for reproducibility
- **Git + GitHub:** version control and collaboration

Components Needed to Build the Product

- **1. Data Ingestion Layer:**
  - **Web Scrapers:** BeautifulSoup, Scrapy
  - **PDF Processor:** PyMuPDF, Tesseract OCR
- **2. Document Processing Layer:**
  - **Text Chunking:** 512 tokens, 56-token overlap
  - **Embedding Model:** text-embedding-3-large (1536-dim)
  - **Vector Database:** Pinecone / ChromaDB
- **3. Retrieval Layer:**
  - **Cosine Similarity Search:** Top-15 candidate retrieval
  - **Cross-Encoder Re-Ranker:** Top-5 final selection

Components Needed to Build the Product

- **4. Response Generation Layer:**
  - **Gemini LLM:** 8K context window, grounded generation
  - **Citation Formatter:** mandatory source attribution
  - **Refusal Mechanism:** declines when context is insufficient
- **5. Frontend and Deployment:**
  - **Streamlit / Next.js:** farmer-facing chat interface
  - **FastAPI + Docker:** backend and containerized deployment

Contents

1. Introduction
2. Motivation
3. Literature Review
4. Gaps in Present Study
5. Feasibility and Scope
6. Risk Identification
7. Requirement Analysis
8. Proposed Methodologies
9. Implementation & Results
10. Conclusion

Implementation Overview

- The implementation follows the five-stage RAG pipeline for agricultural advisory.
- Agricultural data is ingested from ICAR, FAO, KVK, and government sources.
- Document processing is done in two stages:
  - Stage 1: Text extraction, chunking (512 tokens, 56 overlap), metadata storage.
  - Stage 2: Embedding generation (1536-dim) and vector indexing in Pinecone/ChromaDB.
- Final responses are generated using Gemini LLM with mandatory source citations.

Knowledge Base and Data Processing

- Knowledge Base: Agricultural documents from ICAR technical bulletins, FAO guidelines, KVK advisories, and government circulars.
- Core processing steps:
  - web scraping with HTML content extraction and cleaning,
  - PDF text extraction via PyMuPDF and Tesseract OCR,
  - API-based JSON advisory ingestion,
  - Unicode normalization and metadata extraction (source URL, date, organization).
- 512-token chunks with 56-token overlap are generated and embedded using text-embedding-3-large.

Retrieval and Generation Implementation

- Embedding model: text-embedding-3-large producing 1536-dimensional vectors.
- Vector search: Cosine similarity via Pinecone HNSW index, returning Top-15 candidates.
- Re-ranking: Cross-encoder (BERT-based) scores each query-chunk pair, selecting Top-5.
- Generation strategies evaluated:
  - Gemini LLM with 8K token context window,
  - strict system prompt for context-only generation with citations.

Evaluation Protocol and Metrics

- Three retrieval configurations compared:
  - Keyword Only (BM25),
  - Semantic Only (Cosine Similarity),
  - Semantic + Cross-Encoder Re-Ranking (Proposed).
- Primary metrics:
  - Precision@5,
  - Recall@5,
  - F1-Score,
  - Hallucination Rate.
- Additional checks include embedding model comparison, per-stage latency breakdown, and farmer pilot evaluation.

Quantitative Results

- AgriSolve (Semantic + Re-Ranking) outperforms BM25 and Semantic-Only across all metrics.
- Best retrieval performance achieved:
  - Precision@5 = 0.91,
  - Recall@5 = 0.89,
  - F1-Score = 0.90.
- Hallucination comparison:
  - AgriSolve: near-zero hallucination rate,
  - Baseline LLM (Gemini without RAG): 27% hallucination rate.

Latency and Farmer Pilot Evaluation

- End-to-end latency breakdown (avg. over 1,000 queries):
  - Query Embedding: 110ms,
  - Vector Search (Top-15): 450ms,
  - Cross-Encoder Re-Ranking: 180ms,
  - LLM Generation: 1.8–2.5s.
- Total response time: 2.5–3.2 seconds (within farmer-acceptable 3–5s range).
- Farmer pilot (50 participants): 92% found advisory trustworthy, 87% valued source citations, 78% used the tool independently.

Key Findings for Implementation and Results

- Two-stage retrieval (cosine + cross-encoder) improves Precision@5 by 20 percentage points over semantic-only search.
- Context grounding with strict system prompting eliminates hallucination to near-zero levels.
- text-embedding-3-large outperforms MiniLM, BGE, and Nomic models on agricultural domain queries.
- Mandatory citation formatting builds farmer trust and enables advisory verification.
- End-to-end latency of 2.5–3.2s is practical for real-world agricultural use.

Contents

1. Introduction
2. Motivation
3. Literature Review
4. Gaps in Present Study
5. Feasibility and Scope
6. Risk Identification
7. Requirement Analysis
8. Proposed Methodologies
9. Implementation & Results
10. Conclusion

Conclusion

- AgriSolve's five-stage RAG pipeline achieves 91% retrieval precision and near-zero hallucination vs 27% in baseline LLMs.
- Mandatory source citations from ICAR, FAO, and KVK sources ensure verifiable, trustworthy advisory for farmers.
- Future work includes multilingual support, voice interface, image-based disease detection, and offline capability.